ICAD RT-5 · Set 4 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

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2	a	b	c	d
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Numerical — write the value

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Chemistry

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20	a	b	c	d

Numerical — write the value

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Mathematics

1	a	b	c	d
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17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

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22	
23	
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25	

Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 4

Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

Physics

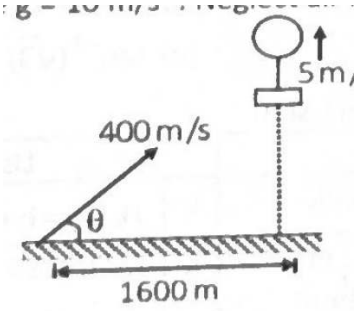
PART A · Q1–20 · SINGLE CORRECT

1. P3L1V5Q4 If point M(1,2,2) moves to N(4,6,3), find |MN|. (a) √35 (b) √26 (c) 7 (d) √29

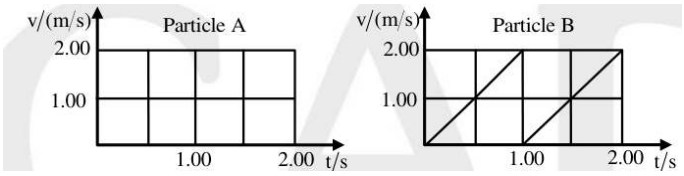
2. P3L2VSuggQ3 The resultant of A+B is R_1. On reversing the vector B, the resultant becomes R_2. What is the value of $R_1^2+R_2^2$? (a) A^2+B^2 (b) A^2-B^2 (c) $2(A^2+B^2)$ (d) $2(A^2-B^2)$

3. P4L2V3Q2 A particle experiences constant acceleration for 20 seconds after starting from rest. If it travels a distance s_1 in the first 10 seconds and distance s_2 in the next 10 seconds, then: (a) $s_2 = s_1$ (b) $s_2 = 2s_1$ (c) $s_2 = 3s_1$ (d) $s_2 = 4s_1$

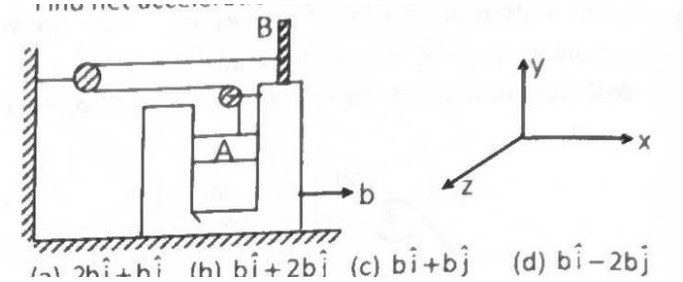
4. P4L9V5Q6 A gun crew observes a remotely controlled balloon launching an instrumented spy package in enemy territory. When first noticed the balloon is at an altitude of 800 m and moving vertically upward at a constant velocity of 5 m/s, and is 1600 m away (horizontally). Shells fired from the gun have an initial velocity of 400 m/s at a fixed angle θ ($\sin\theta = 3/5$ and $\cos\theta = 4/5$). The gun crew waits and fires so as to destroy the balloon. Assume $g = 10\ m/s^2$. Neglect air resistance. How long did the gun crew wait before they fired (in sec)? (a) 10 (b) 40 (c) 50 (d) 80



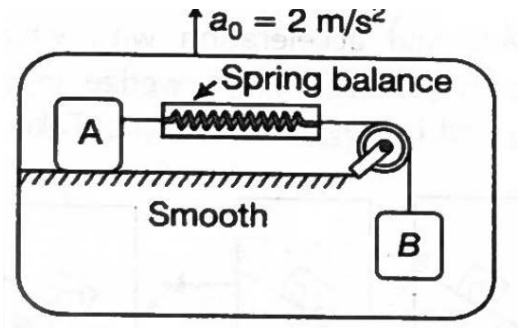
5. P4XVRT2Q17 Two particles A and B start from the same point and move in the positive x-direction. In a time interval of 2.00 s after they start, their velocities v vary with time t as shown in the following figures (particle A moves at 2.00 m/s for the first 1.00 s and at 1.00 m/s for the next 1.00 s; for particle B, v rises uniformly from 0 to 2.00 m/s in each 1.00 s interval). What is the maximum separation between the particles during this time interval? (a) 1.00 m (b) 1.25 m (c) 1.50 m (d) 2.00 m



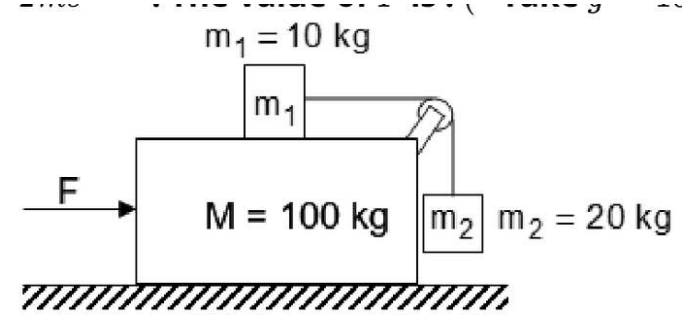
6. P6L3V6Q8 If block B moves towards the right with acceleration b, find the net acceleration of block A. (A slides vertically in a groove of B; the string from B's top post passes round a pulley fixed to the wall, back to a small pulley on B and down to A — see figure.) (a) $2b\hat{i} + b\hat{j}$ (b) $b\hat{i} + 2b\hat{j}$ (c) $b\hat{i} + b\hat{j}$ (d) $b\hat{i} - 2b\hat{j}$



7. P6L7V5Q7 A lift with block system is moving upward with an acceleration 2 m/s². Inside, block A lies on the smooth floor and is tied through a light spring balance and a string over a smooth pulley to block B, which hangs freely. The spring balance is light. The mass of each block is 1 kg. The reading of the spring balance is ($g = 10\ m/s^2$) (a) 1 N (b) 10 N (c) 6 N (d) zero

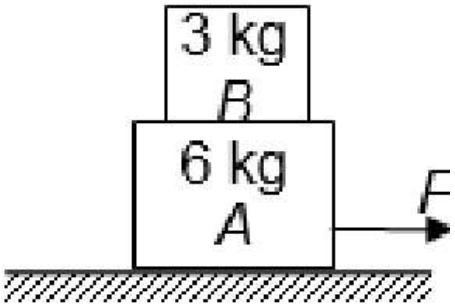


8. P6XVRT3Q9 Three masses M = 100 kg, m_1 = 10 kg and m_2 = 20 kg are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass m_2 moves upward with an acceleration of 2 m s^{-2} (relative to M). The value of F is : (Take g = 10 m s^{-2}) (a) 3360 N (b) 3380 N (c) 3120 N (d) 3240 N



9. P7SBVPYQQ3 A given object takes n times the time to slide down a 45° rough inclined plane as it takes to slide down an identical perfectly smooth 45° inclined plane. The coefficient of kinetic friction between the object and the surface of the inclined plane is (a) $\sqrt{(1 - 1/n^2)}$ (b) $1 - n^2$ (c) $1 - 1/n^2$ (d) $\sqrt{(1 - n^2)}$

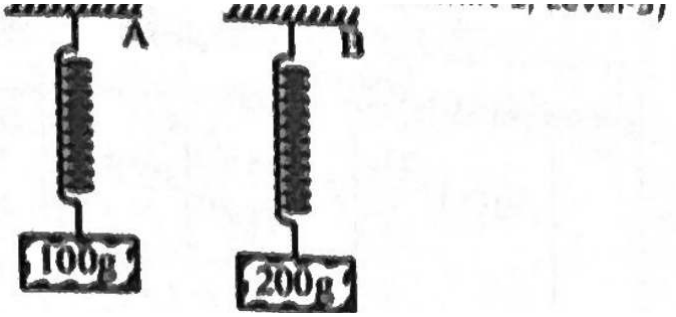
10. P7XVRT4Q10 Two blocks A and B of masses 6 kg and 3 kg rest on a smooth horizontal surface as shown in the figure. If coefficient of friction between A and B is 0.4, the maximum horizontal force which can make them move without separation is (a) 72 N (b) 40 N (c) 36 N (d) 20 N



11. P8L1V5Q4 A force F□ = (5î + 3ĵ + 2k̂) N is applied on a particle which displaces from its origin to the point r□ = (2î – ĵ) m. The work done on the particle in joule is: (a) –7 (b) +7 (c) +10 (d) +13

12. P8L3V3Q4 A bullet when fired at a target with a velocity of 100 m/s penetrates one metre into it. If the bullet is fired at a similar target with a thickness 0.5 metre, then it will emerge from it with a velocity of: (a) 50√2 m/s (b) 50/√2 m/s (c) 50 m/s (d) 10 m/s

13. P8SBVPYQQ5 Two blocks with masses 100 g and 200 g are attached to the ends of springs A and B respectively, as shown in the figure (each spring hangs from the ceiling with its block at the bottom). The energy stored in A is E. The energy stored in B, when the spring constants k A , k B of A and B respectively satisfy the relation 4k A = 3k B , is — (a) 4E (b) 2E (c) 3E (d) (4/3)E — (PYQ)



14. P9L4V5Q6 If a rubber ball falls from a height h and rebounds upto the height of h/2. The percentage loss of total energy of the initial system as well as velocity of ball before it strikes the ground, respectively, are (a) 50%, √(gh/2) (b) 40%, √(2gh) (c) 50%, √(2gh) (d) 50%, √(gh)

PART B · Q21–25 · NUMERICAL VALUE

21. P4XVCAT3Q8 A river of width 500 m has flow velocity 3 m/s. A man starts swimming perpendicular to river flow at a speed of 4 m/s relative to water. Simultaneously a car starts moving on a road, that runs along the opposite side of the river, with constant speed. The car starts from a point on the other side that is exactly opposite to the point from where the man starts swimming. The speed of the car if both, the car and the man, meet at the same point at the moment man has crossed the river, is: (in m/s) (A) 3 (B) 4 (C) 5 (D) none of these

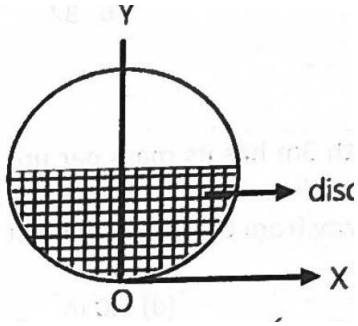
15. P9SBVPYQQ8 A bullet of 10 g leaves the barrel of gun with a velocity of 600 m s –1 . If the barrel of gun is 50 cm long and mass of gun is 3 kg, then value of impulse supplied to the gun will be: (a) 6 N s (b) 3 N s (c) 36 N s (d) 12 N s

16. P10L1V3Q2 An electric fan has blades of length 30 cm as measured from the axis of rotation. If the fan is rotating at 1200 rpm, the acceleration of a point on the tip of the blade is about: ($\pi^2 = 9.86$) (a) 1600 ms –2 (b) 4740 ms –2 (c) 2370 ms –2 (d) 5055 ms –2

17. P10L2VSuggQ1 Which of the following equations represents the angular displacement in circular motion with uniform acceleration? (a) $\omega = \omega_0 + \alpha t$ (b) $\theta = \omega_0 t + (1/2)\alpha t^2$ (c) $\omega^2 = \omega_0^2 + 2\alpha\theta$ (d) $a_c = v^2/r$

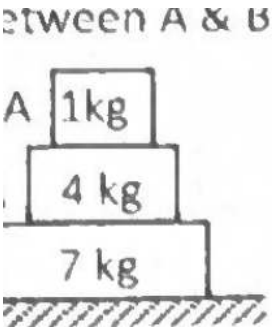
18. P10L3V6Q10 Multiple Correct Answer Question. A car of mass M is moving on a horizontal circular path of radius r. At an instant its speed is v and is increasing at a rate a. (one or more options may be correct) (a) The acceleration of the car is towards the centre of the path. (b) The magnitude of the frictional force on the car is greater than mv^2/r. (c) The friction coefficient between the ground and the car is not less than a/g. (d) The friction coefficient between the ground and the car is $\mu = \tan^{-1}(v^2/rg)$.

19. P11L2V6Q1 A thin uniform semi-circular plate and a semi-circular ring of same mass and radius R are joined as shown (plate forms the lower half, ring the upper half, the X-axis touching the plate at O). The y-coordinate of the centre of mass of the system w.r.t. the shown coordinate system is (a) R(1 + 1/(3π)) (b) R(1 – 1/(3π)) (c) R(π + 1/(3π)) (d) R

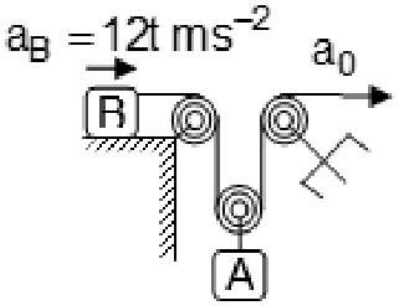


20. P11L4V5Q8 A trolley is moving horizontally with a velocity of v m/s w.r.t. earth (this velocity stays v throughout). A man starts running in the direction of motion of trolley from one end of the trolley with a velocity 1.5v m/s w.r.t. the trolley. After reaching the opposite end, the man turns back and continues running with a velocity of 1.5v m/s w.r.t. trolley in the backward direction. If the length of the trolley is L, then the displacement of the man with respect to earth, measured as a function of time, will attain a maximum value of (a) 4L/3 (b) 2L/3 (c) 5L/3 (d) 1.5L

22. P5L2V4Q4 Three blocks are stacked at rest on the ground: A (1 kg) on top of B (4 kg), B on top of C (7 kg), C on the ground (see figure). Find the normal reaction between A & B in newton.

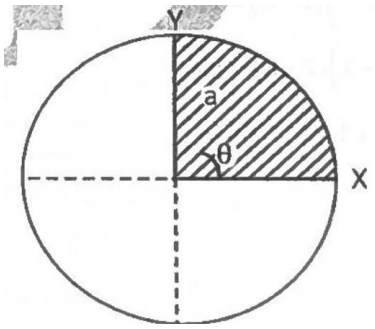


23. P6XVRT3Q25 Block B starts from rest and accelerates as a_B = 12t m s^{-2}. Simultaneously another end of the string is pulled with constant acceleration ao. If at t = 3 s, block A comes to rest then the value of ao (in m s^{-2}) is equal to _____



24. P10XVCAT6Q6 A car is moving on a circular level road of curvature 300 m. If the coefficient of friction between the tyres of the car and road is 0.3, the maximum speed of the car can be _____ (g = 10 m s −2) (A) 81 kmph (B) 90 kmph (C) 162 kmph (D) 108 kmph

25. P11L3V6Q7 The disc of mass M with uniform surface mass density σ is shown in the figure. The centre of mass of the quarter disc (the shaded area) is at the position ((x/3)(a/π), (x/3)(a/π)), where x is _____. (Round off to the nearest integer) (a is the radius of the disc, as shown in the figure)



Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L3VSuggQ4 Two gases A and B at the same temperature and pressure have molar masses 16 and 64 respectively. According to Avogadro's Law and density–molar-mass relations, what is the ratio of their densities (d_A : d_B)? (a) 1 : 4 (b) 4 : 1 (c) 2 : 1 (d) 1 : 2

2. C2L2V6Q3 An atom absorbs a photon of 355 nm and later emits two photons. If one emitted photon has wavelength 680 nm, determine the wavelength of the second photon. Constants: h = 6.626×10^{-34} J s, c = 3.0×10^{8} m s^{-1} (a) 1035 nm (b) 325 nm (c) 743 nm (d) 518 nm

3. C2L5VSuggQ4 Which of the following correctly explains why ionization energy, separation energy and excitation energy are always positive? (a) Electron already has positive total energy in all states (b) Energy must always be supplied to increase the energy of a bound electron (c) Electron releases energy when it moves to higher energy levels (d) Total energy is always positive in all stationary orbits

4. C3L6VSuggQ5 For the reaction 2M + N → product, a chemist mixes 6 mol of M with 4 mol of N, then adds 2 mol more of N (M remains 6 mol). What happens to the amount of product formed? (a) Product increases because N was limiting and now is excess (b) Product remains unchanged because M is the limiting reagent (c) Product doubles because N increases (d) Product becomes zero because M is now excess

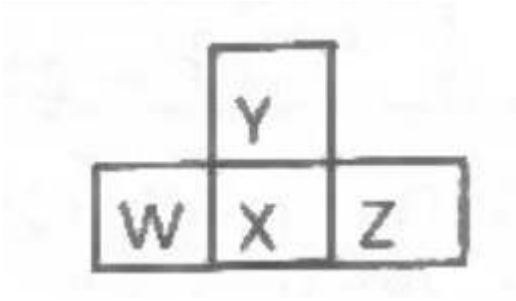
5. C3L10V5Q5 Mole fraction of a solute in an aqueous solution is 0.2. The molality of the solution will be (13.88 / 1.388 / 0.138 / 0.0138)

6. C3XVCAT4Q43 Incorrect statement(s) for 2 M $MgCl_2$ (aq), d = 1.09 g/mL (M $MgCl_2$ = 95): (a) molality of Cl^- is 4.44 m (b) mole fraction of $MgCl_2$ is exactly 0.035 (c) conc. is 19% w/v (d) conc. is 9×10^4 ppm (A) (c) & (d) (B) (b) & (d) (C) (a) & (b) (D) (b) & (c)

7. C4L1V6Q8 Which of the elements whose atomic numbers are given below cannot be accommodated in the present set-up of the long form of the periodic table? (a) 107 (b) 118 (c) 126 (d) 102

8. C4L2V6Q5 For a valence electron in the 3p orbital, which set of electrons contributes 1.00 each to the shielding constant? (a) Electrons in the 3d group (b) Electrons in the 1s group (c) Electrons in the (n−1) shell (d) Electrons to the right of 3p in its group

9. C4L6V6Q3 Consider the following four elements, which are represented according to the long form of periodic table (Y directly above X; W to the left and Z to the right of X). Here W, Y and Z are left, up and right elements with respect to the element 'X', and 'X' belongs to the 16th group and 3rd period. Then according to given information the incorrect statement regarding given elements is: (a) Maximum electronegativity : Y (b) Maximum catenation property : X (c) Maximum electron affinity : Z (d) Y exhibits variable covalency



10. C5L7V3Q3 Why do antibonding molecular orbitals destabilize a molecule? (a) They increase nuclear charge (b) They increase electron density between nuclei (c) They decrease electron density between nuclei (d) They increase overlap strength

11. C5L10V5Q9 The order of strength of hydrogen bonds is: (a) ClH…Cl > NH…N > OH…O > FH…F (b) ClH…Cl < NH…N < OH…O < FH…F (c) ClH…Cl < NH…N > OH…O > FH…F (d) ClH…Cl < NH…N < OH…O > FH…F

12. C5SBVPYQQ112 Number of lone pair(s) of electrons on the central atom and the shape of BrF 3 molecule respectively, are: (a) 0, triangular planar (b) 1, pyramidal (c) 2, bent T-shape (d) 1, bent T-shape

13. C5XVCAT5Q44 Hybridisation involves the mixing of orbitals having comparable energies of same atom. Hybridised orbitals perform efficient overlapping than overlapping by pure s, p or d orbitals. Consider the following compounds and select the incorrect statement from the following: NH_3, PH_3, H_2S, SO_2, SO_3, BF_3, PCl_3, IF_7, P_4, H_2 (a) Six molecules out of given compounds involves hybridization (b) Three molecules are hypervalent compounds (c) Six molecules out of above compounds are non-planar in structure (d) Two molecules out of given compounds involves (dπ–pπ) bonding as well as also involves (pπ–pπ) bonding

14. C6L2V4Q6 Which one is true from the following for isobaric process? (a) $\Delta P = 0$ (b) $\Delta q = 0$ (c) $\Delta H = 0$ (d) $\Delta U = 0$

15. C6L4V5Q1 For the reaction: $FeCO_3(s) \rightarrow FeO(s) + CO_2(g)$; ΔH = 82.8 kJ at 25°C. What is ΔE (or ΔU) at 25°C? (a) 82.8 kJ (b) 80.32 kJ (c) −2394.77 kJ (d) 85.28 kJ

16. C6L6V3Q5 In a bomb calorimeter experiment, 1 g of a substance gives out 4200 J of heat and raises water temperature from 20°C to 30°C. If total water mass is 100 g, find specific heat. (a) 4.2 J/g·K (b) 3.6 J/g·K (c) 4.0 J/g·K (d) 2.1 J/g·K

17. C6L7V3Q4 Given ΔH f : NaCl = −411.15 kJ, HCl = −92.31 kJ, calculate ΔH for Na(s) + HCl(g) → NaCl(s) + (1/2)H 2 (g). Assume ΔH f (H 2) = 0. (a) −318.84 kJ (b) −502.15 kJ (c) −411.15 kJ (d) +92.31 kJ

18. C6L8V3Q3 What is ΔH f ° if U = −788.4, S = 108.5, D = 243.0, IE = 495.8, Δ eg H = −348.6? (a) −788.4 (b) 624.3 (c) −933.1 (d) −411.2

19. C6L9V6Q5 For complete combustion of ethanol, $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$, the amount of heat produced as measured in bomb calorimeter, is 1364.47 kJ mol^{-1} at 25°C. Assuming ideality the Enthalpy of combustion, $\Delta_c H$, for the reaction will be: (R = 8.314 J K^{-1} mol^{-1}) (a) −1366.95 kJ mol^{-1} (b) −1361.95 kJ mol^{-1} (c) −1460.50 kJ mol^{-1} (d) −1350.50 kJ mol^{-1}

20. C6SBVPYQQ22 Which of the following mixing of 1 M base and 1 M acid leads to the largest increase in temperature? (a) 30 mL CH 3 COOH and 30 mL NaOH (b) 45 mL CH 3 COOH and 25 mL NaOH (c) 30 mL HCl and 30 mL NaOH (d) 50 mL HCl and 20 mL NaOH

PART B · Q21–25 · NUMERICAL VALUE

21. C1XVRT1Q49 1 mole of a diatomic gas present in a 10 L vessel at a certain temperature exerts a pressure of 0.96 atm. Under similar conditions an ideal gas exerted 1.0 atm pressure. If the volume of gas molecules is negligible, find the value of van der Waals constant 'a' (in atm·L^2/mol^2).

22. C2XVRT1Q38 A monochromatic yellow radiation of wavelength 595.2 nm is incident on a photosensitive surface. If the total energy of the incident beam is 2.00×10^{-17} J, calculate the number of photons present in the beam. ($h = 6.626\times10^{-34}$ J·s, $c = 3.0\times10^{8}$ m·s^{-1}) (A) 6 (B) 30 (C) 45 (D) 60

23. C4XVRT4Q46 The number of factors that influence the IE are (i) Size of the atom (ii) Charge on the nucleus (iii) Shielding effect (iv) The atomic mass

24. C4XVRT4Q50 How many elements are there in the 6th period of the periodic table?

25. C5SBVPYQQ49 The total number of molecules with zero dipole moment among CH_4, BF_3, H_2O, HF, NH_3, CO_2 and SO_2 is _____

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L2V5Q4 A sector of radius 9 cm has its central angle equal to the exterior angle of a regular n-gon. If the perimeter of the sector is (18 + 6π) cm, then n equals: (a) 3 (b) 4 (c) 6 (d) 9

2. M3XVRT4Q51 If $\sin\theta + \text{cosec}\,\theta = 2$, the value of $\sin^{10}\theta + \text{cosec}^{10}\theta$ is (a) 2 (b) 2^{10} (c) 2^9 (d) 10

3. M4L3VSuggQ3 Find the range of $f(x) = x^2 + 4x + 5$. (a) [1, ∞) (b) (−∞, 5] (c) [0, ∞) (d) [3, ∞)

4. M5AXVRT3Q61 $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$ is equal to (a) 12 (b) 13 (c) 14 (d) 11

5. M5CL4V5Q8 If $a\cos 2\theta + b\sin 2\theta = c$ has α and β as its solutions, then $\tan\alpha + \tan\beta$ is: (a) $(c + a)/2b$ (b) $2b/(c + a)$ (c) $(2 - a)/2b$ (d) $b/(c + a)$

6. M6L1VSuggQ2 If $5 < 9$, what will be the inequality after adding 3 to both sides? (a) $8 < 9$ (b) $8 < 12$ (c) $5 < 12$ (d) $3 < 6$

7. M6SBVPYQQ5 The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has: (a) exactly two solutions in $(-\infty, \infty)$ (b) no solution (c) a unique solution in $(-\infty, 1)$ (d) a unique solution in $(-\infty, \infty)$

8. M7L1V6Q7 Let $a>0$, $a\neq 1$. For real $x>0$ solve $a^{(\log_x a)} = x$. (one or more options may be correct) (a) $x = a$ always satisfies (b) For $a > 1$, exactly one positive solution (c) For $a \in (0,1)$, two positive solutions (d) For large a, no positive solution

9. M8AL3V6Q10 If p, q, r are in A.P. and are positive, the roots of the quadratic equation $px^2 + qx + r = 0$ are all real for (a) $|r/p - 7| \geq 4\sqrt{3}$ (b) $|p/r - 7| < 4\sqrt{3}$ (c) All p and r (d) No p and r

10. M8AXVCAT5Q60 Consider an A.P. with first term 'a' and common difference 'd'. Let S_k denote the sum of first k terms. If S_{kx}/S_x is independent of x, then : (a) $a = 2d$ (b) $a = d$ (c) $2a = d$ (d) none of these

PART B · Q21–25 · NUMERICAL VALUE

21. M2SBVPYQQ1 Let S be the set of the first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \geq 2$ and the product of all elements of B is even$\}$ is _____

22. M4XVRT2Q74 If $2f(xy) = (f(x))\,y + (f(y))\,x$ for all $x, y \in R$ and $f(1) = 2$, then $f(3) =$

23. M7SBVPYQQ8 The number of solutions of the equation $\log_{(x+1)}(2x^2+7x+5)+\log_{(2x+5)}(x+1)^2-4=0$, $x>0$, is

11. M8BL2V3Q5 The sum of first five terms of the series $3 + 4(1/2) + 6(3/4) + \ldots\ldots$ will be (a) 39(9/16) (b) 18(3/16) (c) 39(7/16) (d) 13(19/16)

12. M8CL1V5Q4 If the third and seventh terms of an H.P. are 1/7 and 1/15 respectively, then the n th term is (a) $2n + 1$ (b) $1/(2n + 1)$ (c) $2n - 1$ (d) $1/(2n - 1)$

13. M8DL3V4Q6 Find Tn of the following sequence $1 + 5 + 13 + 29 + 61 + \ldots$ (a) $2^{(n)} - 3$ (b) $2^{(n+1)} - 3$ (c) $2^{(n)} + 3$ (d) $2^{(n+1)} + 3$

14. M8DL5V3Q5 If H = 4 and A = 8, find G. (a) 6 (b) 5 (c) $\sqrt{32}$ (d) 7

15. M8DL5V5Q4 Minimum value of $16\cos^2 x + \sec^2 x$ is (where $x \neq (2n + 1)\pi/2$) (a) 4 (b) 18 (c) 10 (d) 8

16. M9L3V1Q4 If $z = a + ib$, then $z \times \bar{z} = ?$ (a) $a^2 + b^2$ (b) $a^2 \cdot b^2$ (c) $2ab$ (d) $(a + b)^2$

17. M9L6VSuggQ1 Polar form of $z = x + iy$ is: (a) $z = r(\cos\theta + i\cdot\sin\theta)$ (b) $z = x + iy$ (c) $z = a + b$ (d) $z = r\theta$

18. M9L7V2Q2 If n is a multiple of 3 then the value of $\omega^n + \omega^{2n} =$ (a) 0 (b) 1 (c) 2 (d) −1

19. M9L7V6Q8 If $x = a + b$, $y = a\omega^2 + b\omega$ and $z = a\omega + b\omega^2$ where ω is a complex cube root of unity then the product of xyz is equal to……. (a) $a^3 + b^3$ (b) $a + b$ (c) $a^2 + b^2$ (d) $a^4 + b^4$

20. M9L7VSuggQ2 What is $\omega \times \omega^2$? (a) 1 (b) ω (c) ω^2 (d) 0

24. M7XVRT4Q74 If $7^{(\log_7(x^2 - 4x + 5))} = x - 1$, then the smaller value of x is:

25. M8BXVRT4Q72 The first term of an infinite G.P is the value of x satisfying the equation $\log_4(4^x - 15) + x - 2 = 0$ and the common ratio is $\cos(2011\pi/3)$. The sum of GP is